CMPG325 – Computer Networks

Technical Report

A VLAN-based network for Boipelo Cinema & Events

Detail	Information
Project	Individual Project 122
Student number	42067812
Client	Boipelo Cinema & Events (Client ID CL-122), Klerksdorp
Tool	Cisco Packet Tracer
Assigned address block	192.168.53.0/24
Repository	github.com/Refilwe31-07/CMPG325-Individual-Project
Date	16 October 2026

Summary

This report covers the design, build and testing of a small switched and routed network for Boipelo Cinema & Events, a venue in Klerksdorp. The client wanted a dependable network that is easy to look after and that keeps its two departments, Cinema Operations and Events Management, logically separate, while still sharing services like DHCP, DNS and internet access.

I built the network in Cisco Packet Tracer with one router, one Cisco switch, a combined DHCP/DNS server and six PCs, all inside the assigned 192.168.53.0/24 block. VLAN 10 and VLAN 20 are configured, the router sub-interfaces for both VLANs are addressed, and ping tests show that devices inside VLAN 10 can reach each other. One test also shows traffic being routed from VLAN 10 to VLAN 20.

The evidence is not complete, though. Two of the three cross-VLAN tests failed, and several items from the brief (device passwords, port security, default routing and access control lists) are not shown in my screenshots yet. Section 7 lists these gaps honestly, along with the documentation and topology problems I found and how I would fix each one.

1. Introduction

Boipelo Cinema & Events runs two quite different operations under one roof. The cinema side sells tickets and looks after customers. The events side handles bookings, coordination and venue administration. Both need a shared internet connection and a few common services, but neither needs to see the other's everyday traffic.

A VLAN design suits that situation well. It separates the two groups logically without buying two physical networks, which keeps the budget down and makes the network easier to manage. This report explains what the client asked for, how I designed the network to meet it, how I configured it, and what the tests showed.

The scope is limited to what can be built and tested in Packet Tracer. Everything described as "configured" or "tested" is backed by the screenshots in sections 4 and 5 and the appendix. Anything else is described as designed or planned.

2. Client requirements

2.1 Client profile

Item	Detail
Organisation	Boipelo Cinema & Events
Industry	Entertainment
Location	Klerksdorp
Client ID	CL-122
Network technology	Cisco Packet Tracer
Assigned addressing block	192.168.53.0/24

2.2 What the client needs

The client's goal is a dependable network that lets people and devices communicate and share data, while keeping its two departments logically apart. In practice that means the network should:

- support communication between staff members and provide reliable connectivity across the whole venue;
- let authorised users and devices share data;
- stay stable enough for day-to-day business, without interruptions;
- be easy to administer, with secure communication between users and devices.

2.3 Departments and applications

Department	VLAN	Staff	Applications
Cinema Operations	10	Ticket sales and booking management, cinema management, customer service	Ticket booking system, sales system, staff email
Events Management	20	Event coordinators, venue administration, event support staff	Event planning software, venue management system

Both departments also share web browsing, cloud storage and file sharing.

2.4 Services, security, budget and growth

Network services. The client listed DNS for name resolution, DHCP for address assignment, VLAN segmentation, internet access, default routing and print services.

Security. The brief asks for restricted access between departments, VLAN segmentation, passwords on network devices, access control through VLAN membership and secure port configuration. The aim is to protect customer and business information and keep network resources available.

Budget. The client wants cost-effective equipment: a Cisco router, a Cisco layer 2 switch, end devices and standard Ethernet cabling.

Future growth. The subnet 192.168.53.128/25 is reserved for a future branch office. The design should also allow more switches and users to be added without rework.

3. Network design

3.1 Physical topology

The network uses a simple star layout around a single switch. A Cisco 2911 router connects to the switch on one link. Also on the switch are a DHCP/DNS server and six PCs: three in Cinema Operations (Cinema PC1 to PC3) and three in Events Management (Events PC4 to PC6). Everything is connected with copper straight-through cable.

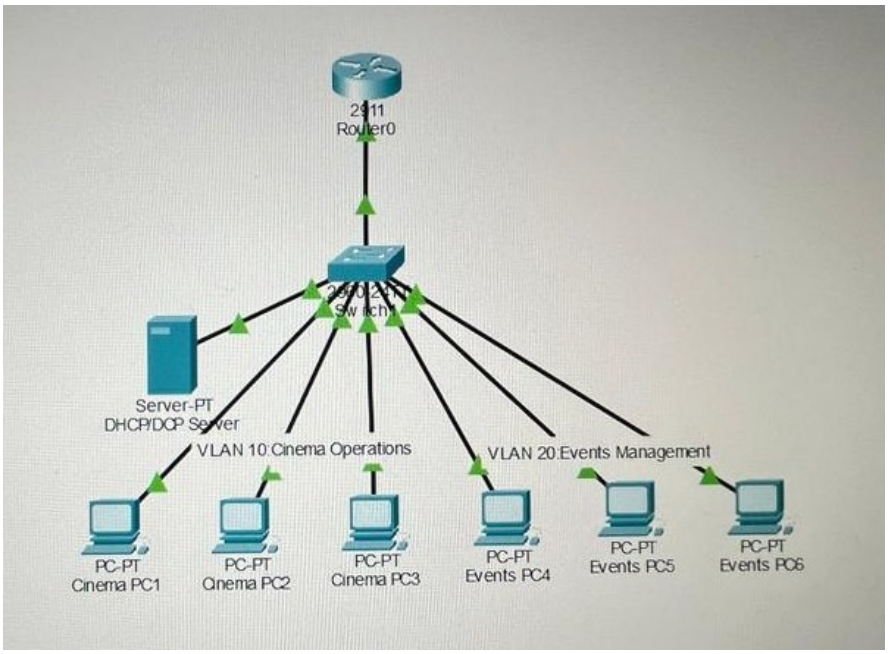


Figure 1: Physical topology in Packet Tracer (2911 router, switch, DHCP/DNS server and six PCs)

In the Packet Tracer view every link shows a green indicator, so all cables are up. The topology contains no internet-facing device and no printer, even though the brief lists both internet access and print services. Those two services cannot be demonstrated until a device for each is added.

3.2 Logical topology and VLANs

The two departments sit in separate VLANs on the same switch. The router connects to the switch through one trunk link and uses a sub-interface for each VLAN to route between them (often called "router-on-a-stick"). Figure 2 redraws the design with the addresses and ports.

VLAN ID	Name	Department	Purpose
10	CINEMA	Cinema Operations	Tickets, administration, customer service
20	EVENTS	Events Management	Event planning, venue management

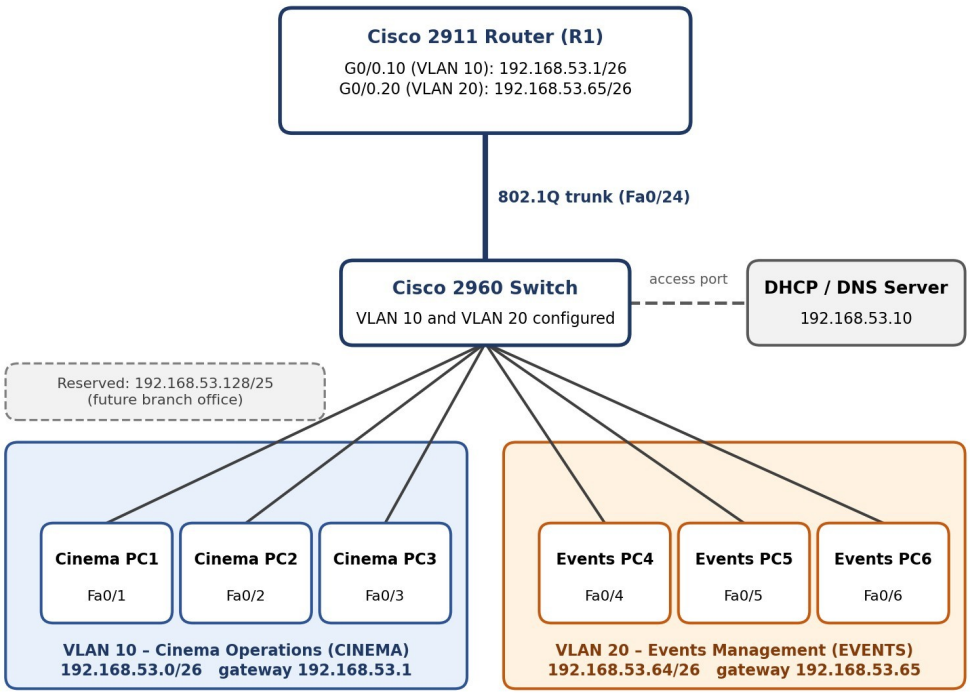


Figure 2: Logical topology showing VLANs, gateways, switch ports and the reserved branch block

3.3 IP addressing plan

The assigned block, 192.168.53.0/24, is split into three parts. Each department gets a /26 (64 addresses, 62 usable), and the top half of the block is held back for the branch office.

Segment	Network	Mask	Gateway	Usable hosts	Broadcast
VLAN 10 Cinema	192.168.53.0/26	255.255.255.192	192.168.53.1	62	192.168.53.63
VLAN 20 Events	192.168.53.64/26	255.255.255.192	192.168.53.65	62	192.168.53.127
Branch reserve	192.168.53.128/25	255.255.255.128	Not assigned	126	192.168.53.255

3.4 DHCP and DNS

A single server at 192.168.53.10 provides both DHCP and DNS. Each VLAN has its own address pool:

VLAN	Network	Default gateway	DNS server	DHCP range
10	192.168.53.0/26	192.168.53.1	192.168.53.10	192.168.53.11 – 192.168.53.62
20	192.168.53.64/26	192.168.53.65	192.168.53.10	192.168.53.66 – 192.168.53.126

Starting the VLAN 10 pool at .11 leaves .2 to .10 free for fixed infrastructure such as the server itself.

3.5 Router interfaces and switch ports

The router needs one sub-interface per VLAN. The table below compares what my design document lists with what the router screenshot actually shows.

Sub-interface	VLAN	Address in design document	Address on the router
G0/0.10	10	192.168.53.1/26	192.168.53.1/26 (confirmed)
G0/0.20	20	192.168.53.5/26	192.168.53.65/26 (confirmed)
G0/0.99	99	192.168.53.9/25	Not visible in the evidence

The two mismatches are explained in section 7. On the switch, the ports are allocated like this:

Port	Device	VLAN / role
Fa0/1 – Fa0/3	Cinema PC1, PC2, PC3	VLAN 10 (access)
Fa0/4 – Fa0/6	Events PC4, PC5, PC6	VLAN 20 (access)
Fa0/24	Router (R1)	Trunk

3.6 Why the design looks like this

- **VLANs instead of two networks.** One switch and one router do the job of two physical networks, which fits the budget and still gives each department its own broadcast domain.
- **/26 subnets.** Sixty-two usable addresses per department is comfortable for a venue this size, and it leaves half the /24 free.
- **Router-on-a-stick.** Inter-VLAN routing needs only one router port and one cable. A layer 3 switch would be faster but costs more than the client wants to spend.
- **Reserved /25.** Holding back 192.168.53.128/25 means the branch office can be added later without renumbering anything.

4. Implementation

4.1 VLANs on the switch

I created VLAN 10 and VLAN 20 on the switch and gave them the names CINEMA and EVENTS. The commands were:

```
Switch>enable
Switch#configure terminal
Switch(config)#vlan 10
Switch(config-vlan)#name CINEMA
Switch(config-vlan)#exit
Switch(config)#vlan 20
Switch(config-vlan)#name EVENTS
Switch(config-vlan)#exit
Switch(config)#end
Switch#show vlan brief
```

At one point I typed show vlan brief while still in configuration mode, and IOS rejected it with an "invalid input" error. The command only works from privileged EXEC mode, so I ended the configuration session and ran it again. The output confirms both VLANs are active, with Fa0/1 to Fa0/3 in VLAN 10 and Fa0/4 to Fa0/6 in VLAN 20.

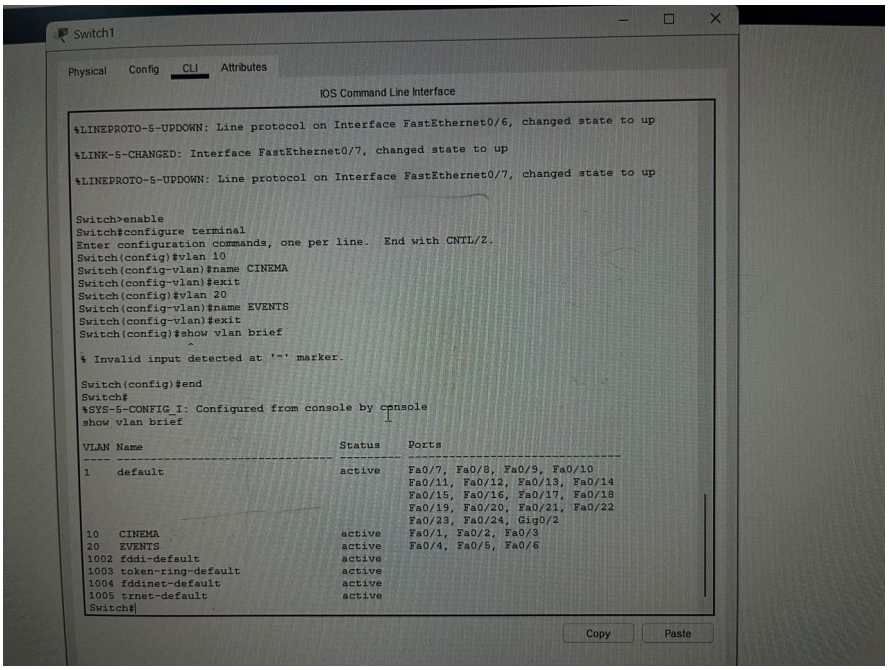


Figure 3: Switch CLI showing VLAN creation and the show vlan brief output

4.2 Router sub-interfaces

The router uses one sub-interface per VLAN, each tagged with 802.1Q. The VLAN 20 sub-interface was configured like this:

```
Router(config)#interface g0/0.20
Router(config-subif)#encapsulation dot1Q 20
Router(config-subif)#ip address 192.168.53.65 255.255.255.192
Router(config-subif)#end
Router#show ip interface brief
```

This was another place where order mattered. When I tried to enter the IP address first, IOS refused it, saying that routing on a LAN sub-interface is only allowed once the sub-interface is part of an 802.1Q VLAN. Setting the encapsulation first fixed it. The second show ip interface brief output shows G0/0.10 at 192.168.53.1 and G0/0.20 at 192.168.53.65, both up.

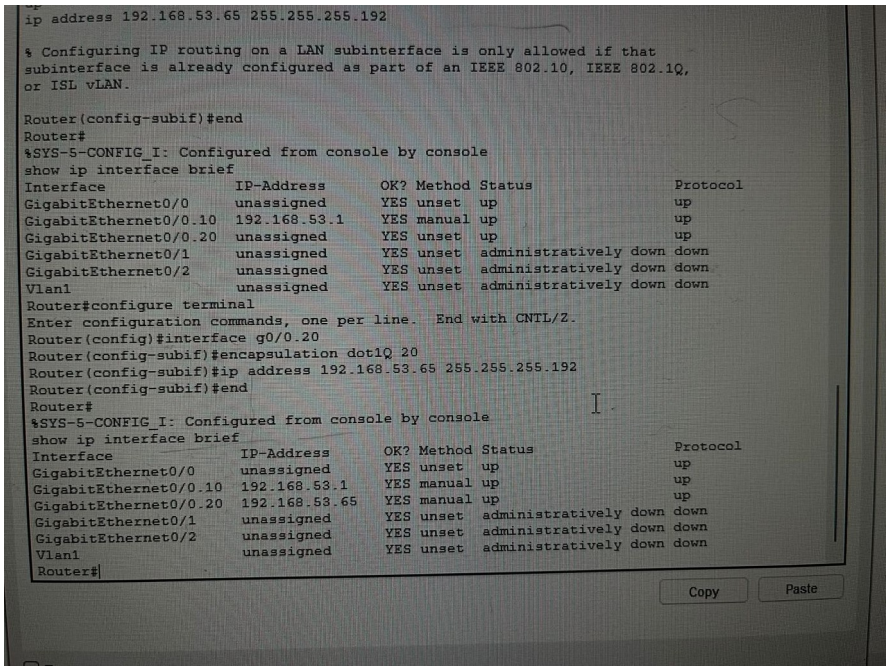


Figure 4: Router CLI showing the sub-interface error and the final show ip interface brief output

4.3 End devices and services

The six PCs are connected to the access ports listed in section 3.5, and the DHCP/DNS server is on the switch at 192.168.53.10. The addresses seen during testing (192.168.53.11 to .13 in VLAN 10, and .70 and .71 in VLAN 20) fall inside the planned DHCP ranges. My screenshots do not show the DHCP pool configuration itself, so I can say the addressing is consistent with the plan but not that I have proved DHCP is handing the addresses out.

5. Testing and results

I tested from the Command Prompt on each Cinema PC, using ping. Each PC first pinged another address in its own VLAN, then tried an address in the Events VLAN. The full screenshots are in the appendix.

Test	From	To	What it checks	Result
1	Cinema PC1	192.168.53.11	Same VLAN	4 of 4 replies, 0% loss, average 5 ms
2	Cinema PC1	192.168.53.70	VLAN 10 to VLAN 20	3 of 4 replies, 25% loss, under 1 ms, TTL 127
3	Cinema PC2	192.168.53.12	Same VLAN	4 of 4 replies, 0% loss, average 5 ms
4	Cinema PC2	192.168.53.71	VLAN 10 to VLAN 20	0 of 4 replies, 100% loss
5	Cinema PC3	192.168.53.13	Same VLAN	4 of 4 replies, 0% loss, average 2 ms
6	Cinema PC3	192.168.53.70	VLAN 10 to VLAN 20	0 of 4 replies, 100% loss

5.1 What the results show

Inside VLAN 10, connectivity is solid. All three same-VLAN tests returned every packet, with round-trip times of a few milliseconds. The TTL of 128 on those replies is the Windows default, which means the packets never crossed a router, as expected for devices in the same subnet.

Routing between VLANs worked once. In test 2 the replies came back with a TTL of 127, one lower than the starting value, so the packets passed through exactly one router. The first request timed out, which is normal while the router and PCs learn each other's MAC addresses, and the next three succeeded. This is good evidence that the 802.1Q sub-interfaces and the trunk to the switch are working.

The other two cross-VLAN tests failed. Tests 4 and 6 lost all four packets. My screenshots do not show why, so I do not want to guess. Possible explanations are that the target host was not yet configured or reachable, that ARP had not completed on a first attempt, or that something is filtering traffic. I have no access list in my evidence, so I cannot call the failure deliberate isolation. These two tests need to be repeated and investigated before I can claim the inter-VLAN behaviour is reliable.

6. Requirements against evidence

The table below checks each requirement from the brief against what my screenshots actually show.

Requirement	Status	Notes
Two departments in separate VLANs	Shown	VLAN 10 and 20 active; ports assigned (Figure 3)
IP addressing per VLAN	Shown	Gateways configured on G0/0.10 and G0/0.20 (Figure 4)
Reliable connectivity inside a department	Shown	Tests 1, 3 and 5
Authorised communication between departments	Partly shown	One of three tests succeeded (test 2)
DHCP	Designed	Pools planned; pool configuration not in the screenshots
DNS	Designed	Server at 192.168.53.10; no name-resolution test shown
Internet access and default routing	Not shown	No WAN link or default route in the evidence
Print services	Not shown	No printer in the topology
Passwords on network devices	Not shown	No password configuration in the evidence
Secure port configuration	Not shown	No port-security commands in the evidence
Restricted access between departments	Not shown	VLAN separation only; no access list shown
Space for a future branch office	Designed	192.168.53.128/25 reserved

7. Issues found and recommended fixes

Going back through the documents against the screenshots, I found several things that need fixing or finishing before the project is complete.

1. **Wrong gateway for VLAN 20 in the router table.** The design lists G0/0.20 as 192.168.53.5/26, but the VLAN 20 plan and the router both use 192.168.53.65. The .5 address would also sit inside the VLAN 10 subnet. The documentation should say 192.168.53.65.
2. **G0/0.99 overlaps the other subnets.** The address 192.168.53.9/25 belongs to the range 192.168.53.0 to .127, which covers both department subnets. There is also no VLAN 99 on the switch. If a management VLAN is wanted, it needs its own small subnet (for example a /28 taken from the

reserved space, such as 192.168.53.240/28) and a matching VLAN on the switch. Otherwise the line should be removed.

3. **VLAN 20 may not reach the DHCP server.** The server is at 192.168.53.10, inside the VLAN 10 subnet. Broadcasts from VLAN 20 do not cross the router, so VLAN 20 clients need a DHCP relay: ip helper-address 192.168.53.10 on the G0/0.20 sub-interface.
4. **Trunk not visible.** In the show vlan brief output, Fa0/24 and Gig0/2 are still listed under the default VLAN. The trunk configuration on Fa0/24 (switchport mode trunk) is not shown. A screenshot of show interfaces trunk would settle this.
5. **Server port and VLAN not documented.** The port allocation table covers only the PCs and the router. The switch log shows Fa0/7 coming up right after Fa0/6, which is probably the server, and show vlan brief lists Fa0/7 under the default VLAN. If that is right, the server is not in VLAN 10 even though its address, 192.168.53.10, belongs to the VLAN 10 subnet, so VLAN 10 clients could not reach it. Assign that port to VLAN 10 (or give the server its own subnet) and add it to the port table.
6. **Label typo and photo quality in the topology.** The server in Figure 1 is labelled "DHCP/DCP Server" and should read DHCP/DNS. The figure is also a photo of the screen, taken at a slight angle. Fixing the label and taking a straight screenshot would make it cleaner.
7. **Missing security configuration.** Device passwords, port security and an access list between VLAN 10 and VLAN 20 are in the brief but not in the evidence. The access list is also what would define "authorised" communication between departments.
8. **Missing services.** Default routing, internet access and print services appear in the brief but not in the build. The topology would need an upstream connection and a printer to demonstrate them.
9. **Repeat the failed tests.** Re-run tests 4 and 6 (ping twice, since the first attempt often loses packets to ARP), and check show ip route on the router and the IP settings on the Events PCs.

8. Future growth

The design leaves room to grow in three ways. The reserved 192.168.53.128/25 block gives the future branch office up to 126 hosts. New departments can be added as further VLANs and sub-interfaces. And because everything hangs off one switch, extra switches can be attached to the trunk as the headcount rises. Each department's /26 also has plenty of spare addresses before it needs resizing.

9. Conclusion

The core of the design is in place and working. Two VLANs separate Cinema Operations from Events Management, the router is configured to route between them, and devices inside VLAN 10 communicate reliably. The addressing plan is sensible, fits the assigned block and leaves space for the branch office.

The build is not yet finished. The cross-VLAN results are inconsistent, and the security and service items in the brief still need to be configured and demonstrated. The fixes in section 7 are all small, and doing them would bring the network in line with what the client asked for.

The project files and configuration are in the GitHub repository:
https://github.com/Refilwe31-07/CMPG325-Individual-Project

Appendix A: Test screenshots

The ping results from section 5, as captured from each Cinema PC.

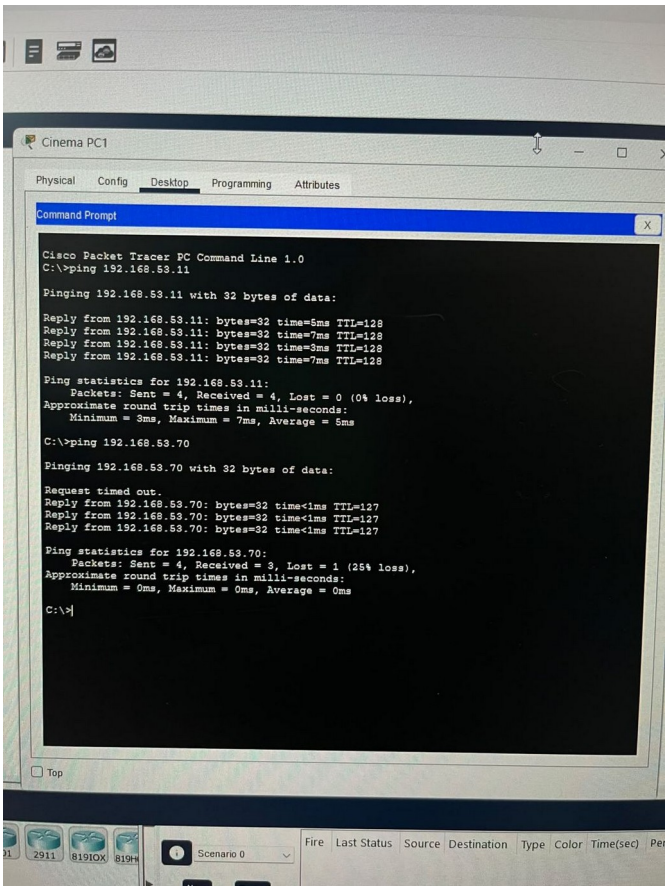


Figure A1: Cinema PC1, ping to 192.168.53.11 (same VLAN) and 192.168.53.70 (VLAN 20)

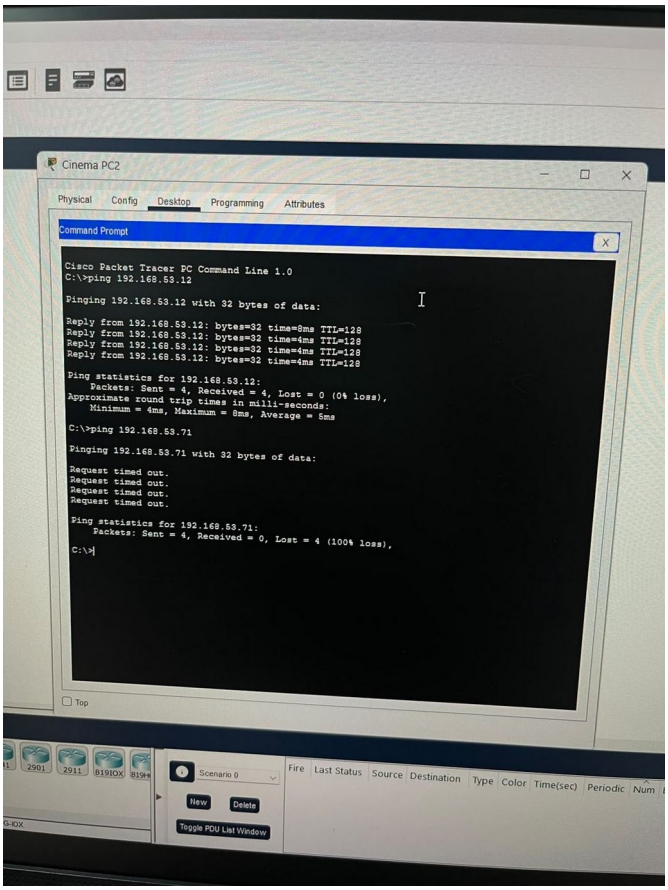


Figure A2: Cinema PC2, ping to 192.168.53.12 (same VLAN) and 192.168.53.71 (VLAN 20)

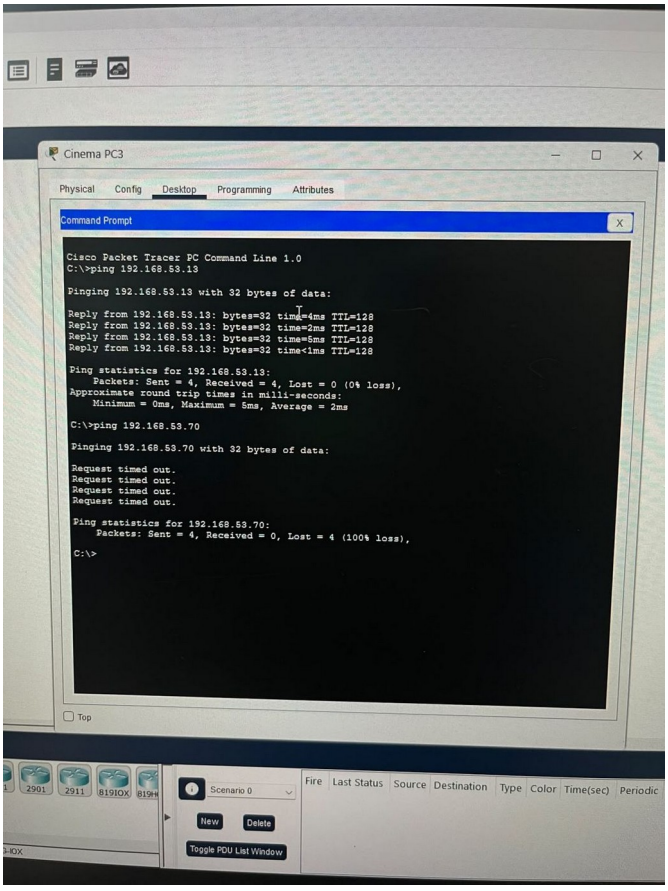


Figure A3: Cinema PC3, ping to 192.168.53.13 (same VLAN) and 192.168.53.70 (VLAN 20)